



GE Medical Systems

Technical Publication

**Direction 5115418-100 Revision 4
Addendum – LS2002M5**

LightSpeed 2.X, 3.X, 4.X

Application Tips and Work-Arounds

LightSpeed QX/i

LightSpeed Plus

LightSpeed Ultra

LightSpeed ¹⁶

HiSpeed QX/i

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CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Healthcare personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Healthcare, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-ray radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-ray radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, its agents (GE Healthcare, GE Medical Systems), and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

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REVISION HISTORY

Revision	Date	Reason for change
1	5/12/2004	Patch C.1 for LS2002M5 2.X, 3.X, 4.X IRIX systems
2	12/8/2004	Patch F for LS2002M5 2X, 3X, 4X IRIX systems
3	1/24/2005	Patch F.2 for LS2002M5 2X, 3X, 4X IRIX systems
4	10/18/2005	Patch G.6 for LS2002M5 2X, 3X, 4X IRIX systems

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ADDENDUM FOR PATCH G.6 LS2002M5 SOFTWARE

SOFTWARE VERSION

Patch G for LS2002M5 – This release of software contains quality fixes to address a number of issues that were reported on the LS2002M4 and LS2002M5 release. This software is to be installed in conjunction with the following releases of software with or without Patch C.1 or Patch F.2: All content for Patch C.1 and Patch F.2 are included in Patch G.6.

LightSpeed 2.X – 2.4.2_H2.4M5

LightSpeed 3.X – 308I.2_H3.1M5

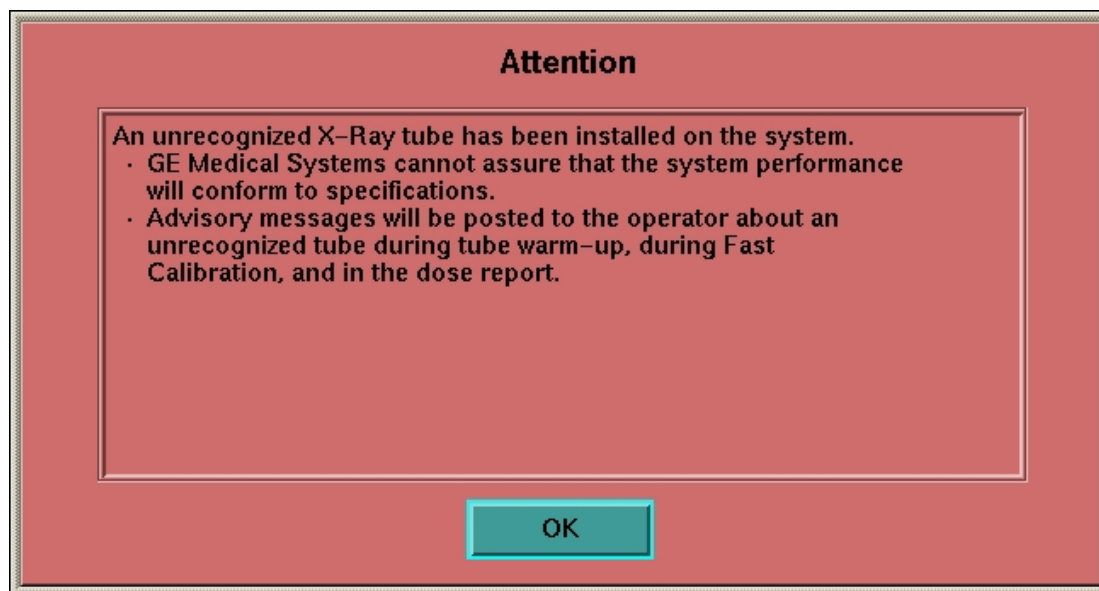
LightSpeed 4.X – 405I.2_H4.0M5 or 405.3_H4.0M5.1

HiSpeed 4X - 101I.2HSQXIM5

SMARTUBE

SmarTube – There are further enhancements to the SmarTube feature in this release. Additional software has been added to indicate identity of the tube based on tube type. For example (1.) GE-Supplied x-ray tube (Performix™) and (2.) all other tube types.

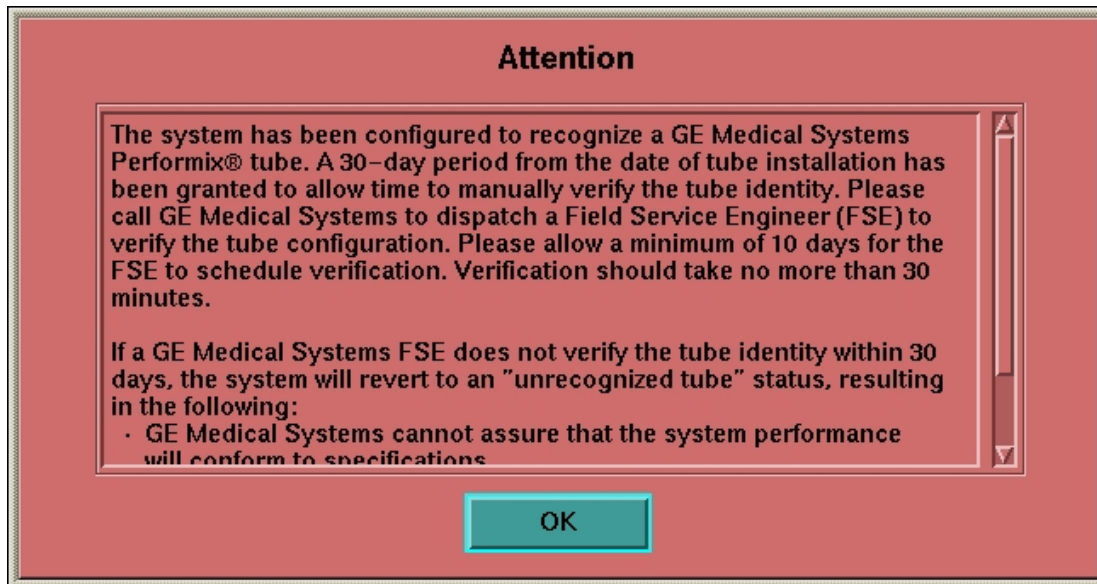
If a Non-GE-Supplied tube is found, the following warning dialog box will be posted at system start-up.



Advisory messages will be posted on systems with Non-GE-Supplied tubes during Fast Cal indicating that the Fast Calibration techniques are designed specifically for GE Medical Systems Performix™ tubes and that the performance of the system cannot be guaranteed with unrecognized tubes. In addition, a message will be posted on the Dose Report screen save image indicating that GE cannot assure the accuracy of the reported dose information for any system configurations that include tubes other than GE Medical Systems Performix™ tubes.

If the installed x-ray tube is a GE-Supplied tube and the field service engineer did not have all the appropriate information to complete verification, a message will be displayed indicating verification needs to be completed.

If the GE-Supplied tube is installed by either an in-house or 3rd party service provider, the same message will be displayed and customer will need to call to have a GE Service Representative dispatched to verify the identity of the tube. The following message will be displayed for 30 days or until the field service engineer is able to complete tube installation verification within the 30-day window.



If the verification is not completed within the 30-day window, software will revert to unrecognized tube state. In that state, advisory messages for an unrecognized x-ray tube will be seen during Start-Up, Tube Warm Up, FastCal and on the Dose Report screen save image. The user will need to acknowledge the messages before he or she can scan.

DOSE REPORT SCREEN SAVE

***Dose Report Screen Save** – feature has been implement in this release. The Dose Report is a screen save image that is automatically saved to Series 999 at normal End Exam. The Dose Report is a copy of the information displayed in the Dose Information Area on the View Edit screen. If an unrecognized tube is in use and additional advisory will be posted at the bottom of the report. In addition, the report includes the CTDIvol, Dose Length Product (DLP) and total DLP for the exam. Scouts will be listed, however, there is no recognized method for measuring dose for scouts so the information for these series will always be blank. If there is more than one group in a series, the series will be listed multiple times showing the dose information per group.

For more information about CTDIvol and DLP, refer to the Dosimetry section in the Technical Reference Manual for your system.

The Dose Report Screen Save can be networked, archived and filmed as part of the patient's record. The Dose Report can be set to be automatically auto filmed in the Film AutoFilm set up page on the View Edit screen.

Patient Name: Sam Ample			Exam no: 507		
Accession Number:			21 Jun 2005		
Patient ID: Sample			LightSpeed Plus		
Exam Description:					
Dose Report					
Series	Type	Scan Range (mm)	CTDIvol (mGy)	DLP (mGy-cm)	Phantom cm
1	Scout	-	-	-	-
2	Helical	S30.000-S7.500	29.75	106.93	Body 32
2	Helical	I72.500-I95.000	29.75	106.93	Body 32
Total Exam DLP:				213.86	
1/1					
W:1 L:-1025					

Dose Report Screen Save for GE x-ray tube

Patient Name: Test Name				Exam no: 504	
Accession Number:				21 Jun 2005	
Patient ID: sfsf				LightSpeed Plus	
Exam Description:					
Dose Report					
Series	Type	Scan Range (mm)	CTDIvol (mGy)	DLP (mGy-cm)	Phantom cm
1	Scout	-	-	-	-
2	Helical	S30.000-S7.500	29.75	106.93	Body 32
Total Exam DLP:				106.93	
Unrecognized tube in use - Dose may vary. The reported dose information is calculated based on empirical observations of systems with GE Medical Systems Performix® tubes. GE cannot assure the accuracy of reported dose information for any configurations that include tubes other than GE Medical Systems Performix tubes.					
1/1					
W:1 L:-1025					

Dose Report Screen Save for non-GE tube

* Note that the Dose Report is available only for IRIX systems with dual processors. Single processors IRIX systems will not have Dose Report enabled. Contact your local GE Sales representative for information regarding an upgrade to Xstream or Xstream FX console that supports Dose Reporting.

ENHANCED TUBE MANAGEMENT

Enhanced Tube Management – feature optimizes rotor usage. This change in rotor usage will help prolong the life of the x-ray tube.

This feature is enabled automatically if the system recognizes that a GE Medical Systems x-ray tube is installed. This feature is a purchased option for Non-GE-Supplied tubes. Contact your local GE sales representative for information.

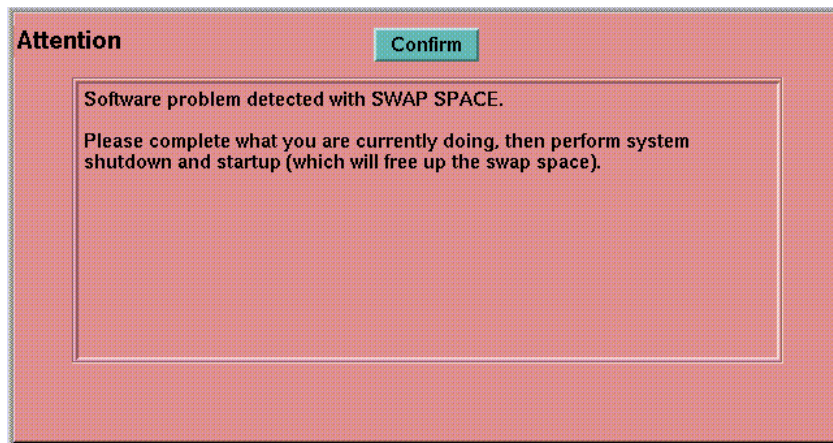
Following message might be seen depending on system operation:

Cannot bring HVDC bus up for xx seconds.

This message informs the system operator that there has been an attempt to enable the system's high voltage bus too soon following a previous successful enabling of the high voltage bus. The operator must wait the specified number of seconds noted in the message before attempting another scan. The HVDC (High Voltage Direct Current) bus timer has been added to the generator management to protect the Power Distribution Unit (PDU) from catastrophic failure resulting from dangerous voltage swings on the HVDC.

SWAP SPACE MESSAGE

Swap Space – If the amount of swap space available is detected to be a low level preventing system from functioning properly, a pop-up message will be displayed to instruct the user to close out actions and reboot the system to recover swap space and system functionality.



DATA LOGGING

Data Logging – enhancements for ProDiags (Prospective Diagnostics) through InSite to include scan usage and tube usage.

DOSE REPORTING

Dose Length Product – issue of incorrect values being reported on the ViewEdit screen when AutomA was turned on and the landmark was lost has been fixed.

FAILED SCOUT RECONSTRUCTION

Scout Reconstruction Failures – the issue where the second scout for the next would fail to reconstruct if recon was still in progress for the last exam has been resolved.

DATA LOG SIZE

Data Logs – there was an issue for some data logs whose size would become so large that they could prevent the system from booting up. Software has been implemented to prevent these logs from growing too large to obstruct system operation.

ADDENDUM FOR PATCH F.2 LS2002M5 SOFTWARE

SOFTWARE VERSION

Patch F.2 for LS2002M5 – This release of software contains quality fixes to address a number of issues that were reported on the LS2002M4 and LS2002M5 release. This software is to be installed in conjunction with the following releases of software with or without Patch C.1:

LightSpeed 2.X – 2.4.2_H2.4M5

LightSpeed 3.X – 308I.2_H3.1M5

LightSpeed 4.X – 405I.2_H4.0M5 or 405.3_H4.0M5.1

HiSpeed 4X - 101I.2HSQXIM5

SMARTUBE

SmarTube – A new feature is implemented in this release. Hardware and software has been added to assure that a GE-Supplied x-ray tube for the system is installed. If a Non-GE-Supplied tube is found, a warning dialog box will be posted at system start-up.

SCOUT RECON FAILURES

Scout Recon Failures – was found to be associated with characters added at the end of the Exam Description when the [Control] or [Shift] key was pressed along with the [Enter] key.

SHOW LOCALIZER ADDED SLICES WHEN TILT PRESCRIBED

Show Localizer with additional slices when tilt prescribed – frequently additional slices were added to graphic prescription when a tilt was prescribed for head scans. Improvements have been made in ScanRx and GraphicRx to address this issue.

SHOW LOCALIZER SLICE LINES INCORRECT

Show Localizer slice lines – were displayed incorrectly if the user moved to [Next Series] quickly skipping a series within the protocol.

GRAPHICALLY PRESCRIBED SLICE LOCATIONS

Graphically prescribed slice locations – adjustments have been made to the Start Locations prescribed in Show Localizer. The Start Location will be rounded to the nearest 0.25mm to provide more productive interface for the users.

NUMBER OF IMAGES IN SCAN PRESCRIPTION

Number of images in scan prescription – may not have been updated correctly due to a rounding error between the Graphic Rx and Scan Rx communications. Number of slices in a prescription is correctly noted and rounding error corrected.

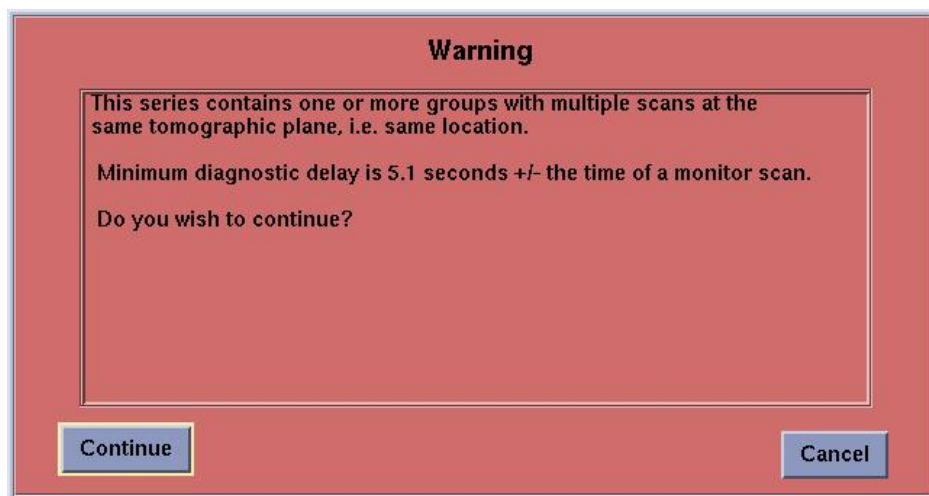
PAGING AND SHOW LOCALIZER

Paging issues connected to Show Localizer – if paging was selected after turning Show Localizer off, paging would fail and user was not able to turn Show Localizer on. This was communication error with hardware and has been corrected.

SMARTPREP

SmartPrep – multiple issues have been addressed in this release.

1. Issues with SmartPrep failing to start were addressed.
2. Communication issues between ScanRx and Display processes were improved to make SmartPrep more reliable.
3. Issues with the SmartPrep diagnostic delay not as expected have been addressed. The diagnostic delay is adjusted at [Confirm] to take into account any table movement that is needed to move from the location for the Monitor scan to the start location for the Scan phase. If the diagnostic delay is going to be longer than what is specified in the SmartPrep menu, the SmartPrep Warning dialog box will display the actual diagnostic delay time to transition from Monitor Phase to Scan Phase.



EXAMRX DISPLAY CRASH

ExamRx Display – a crash which cause the restart of ExamRx Display related to multiple paging requests has been resolved.

RESUME DURING 60 SEC PREP DELAY

Prep delays of greater than 60 sec – improvements have been implemented to better handle long prep delays where the cradle may not be in position to reduce number of resumes previously seen.

STORAGE COMMIT TO PACS

Storage commitment failures – were experienced for some PACS systems due to transfer failures and a dcserver crash. This issue has been resolved.

RESTORE FROM PACS

Restore from PACS – issues were seen when trying to restore images from a PACS that had been flipped or rotated and saved on a RadView system and then sent to the PACS. This issue has been resolved.

Issues restoring helical images from PACS where the item length was unknown has been resolved.

DICOM PRINT OF SCOUT MORE THAN 1200mm

DICOM print of long Scouts – issues were seen when print long Scouts whose format was 1024 instead of 512 in size. Changes made to filming to film long scout as a 512 image if the film format is 5 rows or greater and to film as a 1024 image when number of rows for the film format is 4 rows or less when the long scout is the first image on the page.

AUTOFILM IMAGES FILMED WITH GSE CUTOFF ON FILM

Images filmed with GSE were cut off on film – which was due to issue of filming with GSE on in a non-square view has been resolved.

IMAGES FILMED IN MID INCORRECT

Images filmed in MID mode were flipped, rotated or roamed – root cause was found to be applying GSE and filming in MID using F3 or ShiftF3 to film the viewport. This issue is resolved.

PERFORMED PROCEDURE STEP ID CONFLICT

Performed Procedure Step – issue with conflict of the ID value between the Standard dictionary and Sequence value has been fixed. This issue caused PPS to fail.

DOSE DISPLAY INCORRECT

Dose Display – may have displayed incorrect information if selection of Next Series, Repeat Series, or skipping a series using Next Series was performed quickly. The dose information is now correctly updated.

IMAGE SPACE

Image Space – was reported incorrectly as greater than available for systems where the 3000 Image Series option was not installed or an available as an option. The image space is now correctly reported.

AUTOVOICE FAILED TO PLAY

AutoVoice failed to play – due to a change in the behavior for ISD calculation. This change affected AutoVoice messages of certain lengths. The change that caused the failure of AutoVoice to play was removed.

SECURITY ENHANCEMENTS

Security Enhancements – were added to limit access to a system via FTP, rlogin and telnet to registered IP addresses. This will assure that no unauthorized IP address can access the system.

RELI MONITORING AND PRO DIAGS

Reli Monitoring and Pro Diags – were not working correctly due to an error in the IIP software. The issue has been corrected.

ADDENDUM FOR PATCH C.1 LS2002M5 SOFTWARE

SOFTWARE VERSION

Patch C.1 for LS2002M5 – This release of software contains fixes to address a number of issues that were reported on the LS2002M4 and LS2002M5 release. This software is to be installed in conjunction with the following releases of software:

LightSpeed 2.X – 2.4.2_H2.4M5

LightSpeed 3.X – 308I.2_H3.1M5

LightSpeed 4.X – 405I.2_H4.0M5 or 405.3_H4.0M5.1

HiSpeed 4.X -101I.2HSQXIM5

DISPLAY HANG WITH 10% LOADING MESSAGE

Display Hang – A message box with 10% Loading is displayed and images can no longer be reviewed in the viewport. A system reboot was required to clear the message box and allow display to function in the viewport. This issue has been fixed.

SCOUT RECON FAILURES

Scout Recon Failures – were found for various lengths of scouts. One length that routinely caused recon failures was 175mm. This recon issue has been fixed and scouts will reconstruct without error.

TUBE HEAT OPTIMIZATION WITH AUTOMA

AutomA – a number of issue with tube optimization have been fixed when using AutomA. Interaction of the system calculation of tube heat and maximum mA possible has been improved to provide more consistent system operation.

RECONSTRUCTION SLOW DOWN

Reconstruction Speed – an issue with reconstruction speed for some pitches, detector configurations and intervals less than the detector width has been resolved and reconstruction speeds are the same as in LS2002M4.

GRAPHIC RX PAGE UP/PAGE DOWN

Graphic Rx – issues were seen when using page up and page down and the next or prior scout not being displayed have been addressed. Improvements for display of slice lines were made as well. Instrumentation was done to help trouble shoot issue in Graphic Rx.

SMARTSTEP

SmartStep – multiple issues were addressed. Image out order, which was a system defect where new SmartStep images were not displayed if a display action, was active during acquisition of new data. If an image is not available for display, the viewport will be blank. Using the prior/next buttons will toggle through the images and if the images are available will replace the blank viewports.

AUTOVOICE WITH TABLE MOVE

AutoVoice – AutoVoice message will still play when the table is moved during a long delay. If the table is moved to the correct location, the prescribed AutoVoice message will play automatically. If the table is not moved back to the correct location, the Table Move light will come on. The system will pause the scan, as soon as the table is moved back to the correct location, the scan can be resumed and the prescribed AutoVoice message will be played.

SMARTSCORE

SmartScore – improvements have been implemented to better calculate the heart rate and handle variability in the heart rate during a SmartScore cine acquisition. Also, irregular heart beats (PVCs) will be ignored by the system to improve quality of the SmartScore data.

SYSTEM DISK SPACE

System Disk Space – error in calculation of available space on the system disk for images have been resolved. Depending on the size of the system disk installed on your system, you may see a larger image space available than on previous software releases.

NEW PRECAUTIONS PATCH G.6 FOR LS2002M5

SYSTEM

- For full list of issues refer to the Work-Arounds provided with the LS2002M5 software release.
 - LightSpeed 2.X systems – Direction 2243321, rev 10
 - LightSpeed 3.X systems – Direction 2296447, rev 5
 - LightSpeed 4.X systems – Direction 2352974, rev 3, and HiSpeed QX/I
- End Exam may be slow to complete. At End Exam, the real time statistics are being collected and until this process is complete there may be times when End Exam is slow to complete.
- At End Exam, cursor responses may be slow while the Dose Text Page is being created. Please wait a few seconds if you do not get a response from a mouse action.

AUTOMA

- If Next Series is selected quickly after acquisition of Scout scans when AutomA is enabled for the next series, AutomA may be turned off. AutomA can be enabled for the series.

AUTOFILM

- Auto Film will automatically print the page before printing the Dose Report.

ILINQ

- Accessories web page is not accessible via iLinq. Please use a web connection from a PC.

DOSE TEXT PAGE

- The Swap Space pop-up message is encountered during an exam. The Dose Text Page will not be created at End Exam.
- Dose text Page may not be created if Anonymous Patient is in progress when End Exam is selected and if the disk space is limited.

SPACE AVAILABLE

- Free space on the system maybe less than noted due to Dose Text Pages not being calculated in space available. Maintain adequate free space to support scanning on a daily basis. Remember do not remove image while the system is scanning. Remove images at time when the system is idle.
- Anonymous Patient does not take into account available space. Check available space before creating anonymous data.

NEW PRECAUTIONS PATCH C.1 FOR LS2002M5

SYSTEM

- For full list of issues refer to the Work-Arounds provided with the LS2002M5 software release.
 - LightSpeed 2.X systems – Direction 2243321, rev 10
 - LightSpeed 3.X systems – Direction 2296447, rev 5
 - LightSpeed 4.X systems – Direction 2352974, rev 3, and HiSpeed QX/i
- Annotation may not intermittently be displayed on an image. Move the cursor into the viewport to display the annotation.

SMARTSTEP

- Roam and zoom applied to Scout image in the free viewport are not carried over to the next Scout image. The roam or zoom setting will have to be reapplied.
- Selecting a graphic applied to an image in the free viewport in SmartStep will cause that viewport to be placed in focus with the interventional viewports. Change the focus to the free viewport first by clicking anywhere in the viewport and then select the graphic.
- Screen Save images created in SmartStep are not shown in the Browser until after the SmartStep series is ended.
- Trackball paging in SmartStep in the free viewport may cause interventional viewport to be selected as well. Please do not use trackball paging in SmartStep.
- User may experience issues trying to interact with the mouse while trackball paging is active.
- Loading PROC or COMB image types into the free viewport for SmartStep will cause SmartStep display to exit. Do not select these image types for review in SmartStep.
- If annotation text field is moved to the edge of the viewport, it can become lost. If you no longer desire the annotation text field on the image, erase the text field instead of moving it to the edge of the viewport.

DIRECTVIS

- Film and save for the DirectVis applications of Direct3D and VariViewer may fail if Apply is quickly selected after selecting Film or save. Select Film or Save again.

PERFUSION

- Acquisitions with Extended CT Range On from LightSpeed Pro, LightSpeed RT or 1X and 2X systems with Xstream should not be processed on the OC. The results will be lower than expected compared to data with Extended CT Range off.

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